

Jul 17

Materials to cover: § 2.1 - 2.2.

§ 2.1 Functors of Points

First, some review on (naïve) Category theory.

Def: A category $\mathcal{C}$ consists of

- A class $\text{Ob}(\mathcal{C})$ of objects (sometimes just write $\mathcal{C}$).
- A class $\text{Hom}(\mathcal{C})$ of morphisms / maps / arrows with source and target in $\text{Ob}(\mathcal{C})$.

morphisms from object A to object B are denoted $\text{Hom}_{\mathcal{C}}(A, B)$.

- A binary operation $\circ: \text{Hom}(B, C) \times \text{Hom}(A, B) \rightarrow \text{Hom}(A, C) \quad \forall A, B, C \in \text{Ob}(\mathcal{C})$

called the composition, satisfying ① Association $(f \circ g) \circ h = f \circ (g \circ h)$

② exist identity on each object: $f: A \rightarrow B$,
 $f \circ \text{id}_A = f = \text{id}_B \circ f$.

Example:	Category	Object	Morphisms (also called <u>Cat</u> - maps // maps)
	Set	sets	functions
	Group	groups	group homomorphisms
	Ab	abelian groups	group homomorphisms
	Ring	rings with unity	ring homomorphism
fix k	k -Vec	k -vector spaces	k -linear transformations
fix R	R -mod	(left) R -modules	R -linear maps
	Top	topological spaces	continuous maps
	Sch	k -schemes of f.t.	k -scheme maps (See Chi Ki's notes)

Basically, the morphisms are maps with certain properties that

- make sense in the class of objects, and
- preserve certain structure of the class of objects

One more example :

$\mathcal{C}$ is a category, $\mathcal{C}^{op}$ the opposite category :

$$Ob(\mathcal{C}^{op}) = Ob(\mathcal{C})$$

Morphisms are in opposite direction.

$$Hom_{\mathcal{C}}(X, Y) \xrightarrow{\sim} Hom_{\mathcal{C}^{op}}(Y, X)$$

$$f \mapsto f^{op}$$

$$\text{and } f^{op} \circ g^{op} = (g \circ f)^{op}.$$

Def: A (covariant / contravariant) functor is a morphism between categories.

$F: \mathcal{C} \longrightarrow \mathcal{D}$ includes

• $\forall C \in \mathcal{C}$, get $F(C) \in \mathcal{D}$.

• $\forall f: C \longrightarrow C'$ in $Hom_{\mathcal{C}}(C, C')$, get

(covariant :) $F(f): F(C) \longrightarrow F(C')$ in $Hom_{\mathcal{D}}(F(C), F(C'))$

(contravariant :) $F(f): F(C) \longleftarrow F(C')$ in $Hom_{\mathcal{D}}(F(C'), F(C))$

Agreeing with compositions and identity.

(covariant :) $F(f \circ g) = F(f) \circ F(g)$

(contravariant :) $F(f \circ g) = F(g) \circ F(f)$.

$$F(id_X) = id_{F(X)}.$$

Example : ① $\mathcal{C} = \text{Ring}, \text{Ab}, R\text{-Mod}, \dots$

$$\mathcal{D} = \text{Set}.$$

$F = \text{forgetful functor}.$

② $\mathcal{C} = \text{any cat.}$

$$X \in Ob(\mathcal{C})$$

$Hom(X, -)$ is a covariant functor $\mathcal{C} \longrightarrow \text{Set}$

$$Y \mapsto Hom(X, Y)$$

$$(f: Y \rightarrow Z) \mapsto \left(F(f): Hom(X, Y) \rightarrow Hom(X, Z) \right)$$
$$g \mapsto f \circ g$$

③ $\text{Hom}(-, X)$ is a contravariant functor $\mathcal{C} \rightarrow \text{Set}$

$$Y \mapsto \text{Hom}(Y, X)$$

$$(f: (Y \rightarrow Z)) \mapsto \left(\begin{array}{c} F(f): \text{Hom}(Z, X) \rightarrow \text{Hom}(Y, X) \\ g \mapsto g \circ f \end{array} \right)$$

Def: Natural transformations (morphism between functors) (It is the correct notion of maps in the category $\text{Fun}(\mathcal{C}, \mathcal{D})$)

Let $F, G: \mathcal{C} \rightarrow \mathcal{D}$ be functors.

A natural transformation $\eta: F \rightarrow G$ consists of the data

$$\cdot \forall X \in \mathcal{C}, \eta_X: F(X) \rightarrow G(X)$$

$$\cdot \forall f: X \rightarrow Y, \quad \begin{array}{ccc} F(X) & \xrightarrow{\eta_X} & G(X) \\ F(f) \downarrow & \circ & \downarrow G(f) \\ F(Y) & \xrightarrow{\eta_Y} & G(Y) \end{array} \quad \text{if contravariant}$$

If η_X is an isomorphism for all $X \in \mathcal{C}$, we call η a natural isomorphism or an isomorphism of functors.

Def: The functor of points

The functor of points of a scheme X is the example ③ above, with $\mathcal{C} = \text{Sch}$.

$$\left(\begin{array}{l} h_X := \text{Hom}(-, X) : \text{Sch} \rightarrow \text{Set} \\ \forall Y \in \text{Sch}, h_X(Y) := \text{Hom}(Y, X) \\ \forall Y, Z, \forall f: Y \rightarrow Z, h_X(f): h_X(Z) \rightarrow h_X(Y), g \mapsto g \circ f. \end{array} \right)$$

Furthermore, a morphism of schemes $f: X \rightarrow Y$ induces natural transformation

$$\eta_f: h_X \rightarrow h_Y,$$

$$\text{where } \forall Z \in \text{Sch}, \eta_{f, Z}: \underset{\text{Hom}(Z, X)}{h_X(Z)} \rightarrow \underset{\text{Hom}(Z, Y)}{h_Y(Z)}$$

$$g \mapsto f \circ g.$$

(precomposing).

Def: Category of presheaves on Sch

$$\cdot \text{Psh}(\text{Sch}) := \text{Fun}(\text{Sch}^{\text{op}}, \text{Set})$$

(i.e. objects are contravariant functors from Sch to Set, called presheaves.)
morphisms are natural transformations.

• The functor of points $h_x = \text{Hom}(-, x)$ is a contravariant functor from Sch to Set,
so $h_x \in \text{Psh}(\text{Sch})$.

For any $X \in \text{Sch}$, we have h_x , so in fact we have a functor

$$h: \text{Sch} \longrightarrow \text{Psh}(\text{Sch})$$

$$X \longmapsto h_x = \text{Hom}(-, x)$$

$$(f: x \rightarrow y) \longmapsto h_f: h_x \longrightarrow h_y$$

$$(g \in \text{Hom}(-, x) \longmapsto f \circ g: \text{Hom}(-, y))$$

• $\text{Psh}(\mathcal{C})$ works for any category $\mathcal{C}$.

Example (2.3)

$$\begin{aligned} \cdot h_x(\text{Spec } k) &= \text{Hom}(\text{Spec } k, X) \\ &= \text{set of } k\text{-points of } X. \end{aligned}$$

$$\begin{aligned} \cdot h_{A'}(Y) &= \text{Hom}(Y, A') \\ &= \text{set of functions on } Y \text{ (morphisms } Y \rightarrow A') \end{aligned}$$

$$\begin{aligned} \cdot h_{\mathbb{G}_m}(Y) &= \text{Hom}(Y, \mathbb{G}_m) \\ &= \text{set of invertible functions on } Y. \end{aligned} \quad \begin{array}{l} \text{(scheme)} \\ (\mathbb{G}_m = A' \setminus \{0\}, \text{ multiplicative group}) \end{array}$$

Lemma 2.4: (Yoneda Lemma)

Let $\mathcal{C}$ be any category.

Then $\forall X \in \mathcal{C}, \forall F \in \text{Psh}(\mathcal{C})$, there is a bijection

$$\{\text{natural transformations } \eta: h_x \longrightarrow F\} \longleftrightarrow F(X)$$

given by

$$\eta \longmapsto \eta_x(\text{id}_x)$$

(Proof omitted, just show bijection. Injectivity need a bit diagram chasing.)

Pf: Surjective :

$\forall s \in F(X)$, want η s.t. $\eta_X(id_X) = s$.

For $Y \in \mathcal{C}$, let $\eta_Y : h_X(Y) \rightarrow F(Y)$ be the morphism of sets
 $(f: Y \rightarrow X) \mapsto \overline{F(f)(s)}$ in $F(Y)$.

$F(X) \rightarrow F(Y)$ since F is a functor.

Then form natural transformation $\eta : h_X \rightarrow F$.

Check $\eta_X : h_X(X) \rightarrow F(X)$,

$$id_X \mapsto \underbrace{F(id_X)(s)}_{id_{F(X)}} = s$$

Injective :

Suppose $\eta_X(id_X) = \eta'_X(id_X)$, $\eta, \eta' : h_X \rightarrow F$ natural transformations.

Then $\forall Y \in \mathcal{C}$, $\forall g: Y \rightarrow X$, want $\eta_Y(g) = \eta'_Y(g)$.

$$\begin{array}{ccc} h_X(X) & \xrightarrow{\eta_X} & F(X) \\ h_X(g) \downarrow & & \downarrow F(g) \\ h_X(Y) & \xrightarrow{\eta_Y} & h_X(Y) \end{array} \quad \begin{array}{ccc} h_X(X) & \xrightarrow{\eta'_X} & F(X) \\ h_X(g) \downarrow & & \downarrow F(g) \\ h_X(Y) & \xrightarrow{\eta'_Y} & h_X(Y) \end{array}$$

$$\Rightarrow \eta_Y(g) = (\eta_Y \circ h_X(g))(id_X) = (F(g) \circ \eta_X)(id_X)$$

$$= F(g)(\eta_X(id_X))$$

$$= F(g)(\eta'_X(id_X))$$

$$= (\eta'_Y \circ h_X(g))(id_X) = \eta'_Y(g)$$

$$\eta_Y : h_X(Y) \rightarrow F(Y)$$

The functor $h : \mathcal{C} \rightarrow \text{Psh}(\mathcal{C})$ is called Yoneda embedding.

Corollary (2.5) :

$h : \mathcal{C} \rightarrow \text{Psh}(\mathcal{C})$ is fully faithful, i.e.

$\text{Hom}_{\mathcal{C}}(X, Y) \longrightarrow \text{Hom}_{\text{Psh}(\mathcal{C})}(h_X, h_Y)$ is bijective.

Definition: (Representable functors)

A presheaf $F \in \text{Psh}(\mathcal{C})$ is called representable if $\exists X \in \mathcal{C}$

s.t. $F \cong h_X$ (natural isomorphism).

(i.e. F is isomorphic to an image of the Yoneda embedding.)

Not all presheaves $F \in \text{Psh}(\text{Sch})$ are representable by a scheme X !

We are interested in 'moduli functors'.

Bigger picture: Moduli problems.

→ We want to classify a collection of objects up to equivalence.

→ Even more, we want structure (moduli space) that encodes how these objects vary

(continuously in families.

→ This info is encoded in moduli functors.

Def: Naive moduli problem: $(\mathcal{A}, \sim)$

a collection $\mathcal{A}$ of objects with an equivalence relation $\sim$ on $\mathcal{A}$.

Examples:

① $\mathcal{A} = \text{Gr}(k, n)$: k -dim lin. subsp. of $\mathbb{A}^n$, $\sim$ is equality

② $\mathcal{A} = \mathcal{M}_{0,n}$: set of n ordered, distinct points on $\mathbb{P}^1$, $\sim$ is action by PGL_2 (automorphism group) on $\mathbb{P}^1$.

③ $\mathcal{A}$ = set of hypersurfaces of degree d in $\mathbb{P}^n$, $\sim$ is equality or PGL_{n+1} action.

④ $\mathcal{A}$ = collection of vector bundles on a fixed scheme X , $\sim$ is isom of V.B.

⚡ These are, a priori, sets. Where is the structure we want?

Aim: Find a scheme M whose $\mathbb{A}^1$ -points are in bijection with the set of equivalence classes $\mathcal{A}/\sim$.

We also want M to encode how these objects vary continuously in 'families'.

(get family over S by pullback of morphism.)

Our aim is to find a scheme M whose k -points are in bijection with the set of equivalence classes $\mathcal{A}/\sim$. Furthermore, we want M to also encode how these objects vary continuously in 'families'. More precisely, we refer to $(\mathcal{A}, \sim)$ as a naive moduli problem, because there is often a natural notion of families of objects over a scheme S and an extension of $\sim$ to families over S , such that we can pullback families by morphisms $T \rightarrow S$.

Definition 2.10. Let $(\mathcal{A}, \sim)$ be a naive moduli problem. Then an extended moduli problem is given by

- (1) sets $\mathcal{A}_S$ of families over S and an equivalence relation $\sim_S$ on $\mathcal{A}_S$, for all schemes S ,
- (2) pullback maps $f^* : \mathcal{A}_S \rightarrow \mathcal{A}_T$, for every morphism of schemes $T \rightarrow S$,

satisfying the following properties:

- (i) $(\mathcal{A}_{\text{Spec } k}, \sim_{\text{Spec } k}) = (\mathcal{A}, \sim)$;
- (ii) for the identity $\text{Id} : S \rightarrow S$ and any family $\mathcal{F}$ over S , we have $\text{Id}^* \mathcal{F} = \mathcal{F}$;
- (iii) for a morphism $f : T \rightarrow S$ and equivalent families $\mathcal{F} \sim_S \mathcal{G}$ over S , we have $f^* \mathcal{F} \sim_T f^* \mathcal{G}$;
- (iv) for morphisms $f : T \rightarrow S$ and $g : S \rightarrow R$, and a family $\mathcal{F}$ over R , we have an equivalence $(g \circ f)^* \mathcal{F} \sim_T f^* g^* \mathcal{F}$.

For a family $\mathcal{F}$ over S and a point $s : \text{Spec } k \rightarrow S$, we write $\mathcal{F}_s := s^* \mathcal{F}$ to denote the corresponding family over $\text{Spec } k$.

Def 2.10 Rewrite :

Let $(\mathcal{A}, \sim)$ be a naive moduli problem.

An extended moduli problem is given by

- $\forall$ scheme S , $\mathcal{A}_S$ is a set of families over S , with eqv. relation $\sim_S$.
- $\forall$ morphism of schemes $f : T \rightarrow S$, get pullback map $f^* : \mathcal{A}_S \rightarrow \mathcal{A}_T$,

satisfying :

$$(i) (\mathcal{A}_{\text{Spec } k}, \sim_{\text{Spec } k}) = (\mathcal{A}, \sim)$$

(k -points are in bijection with the set of equivalence classes of $(\mathcal{A}, \sim)$).

$$\left[\begin{array}{l} \text{Functor} \\ \text{conditions /} \\ \text{nice} \\ \text{properties} \end{array} \right. \left[\begin{array}{l} (ii) \forall \text{ family } \mathcal{F} \text{ over } S, \text{Id}_S^* \mathcal{F} = \mathcal{F}. \\ (iii) \text{ If } \mathcal{F} \sim_S \mathcal{G} \text{ over } S, \text{ then } f^* \mathcal{F} \sim_T f^* \mathcal{G} \text{ over } T. \\ (iv) f : T \rightarrow S, g : S \rightarrow R, \mathcal{F} \text{ family over } R, \text{ then} \\ \quad (g \circ f)^* \mathcal{F} \sim_T f^* (g^* \mathcal{F}) \text{ over } T. \end{array} \right]$$

$T \xleftarrow{\text{example: If } T = \text{Spec } k, \text{ then } \mathcal{F}_s \sim \mathcal{G}_s.}$
 $\downarrow$
 S
 $\downarrow$
 $\text{Spec } k$ (fiber of different family, eqv in $\sim$)

Notation :

For a family $\mathcal{F}$ over S and a point $s : \text{Spec } k \rightarrow S$,

write $\mathcal{F}_s := s^* \mathcal{F}$ to denote the corresponding family over $\text{Spec } k$.

Def/Lemma 2.11 : Moduli functor .

Fun(Sch^{op}, Set)

An extended moduli problem (from $(\mathcal{A}, \sim)$) defines a functor $\mathcal{M} \in \text{Psh}(\text{Sch})$ given by

$$\mathcal{M}(S) := \{\text{families over } S\} / \sim_S$$

$$\mathcal{M}(f: T \rightarrow S) = f^*: \mathcal{M}(S) \rightarrow \mathcal{M}(T)$$

Note: there can be different extensions of a naive moduli problem.

Example (2.13):

Möbius transform by $\begin{bmatrix} a & b \\ c & d \end{bmatrix} \in \text{PGL}_2$:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \end{bmatrix}$$

Example: $p_1 = [1:1]$, $p_2 = [2:1]$, $p_3 = [3:1]$

$$\begin{bmatrix} 1 & -1 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \end{bmatrix} = [0:1] \quad "0"$$

$$\begin{bmatrix} 1 & -1 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} = [1:1] \quad "1"$$

$$\begin{bmatrix} 1 & -1 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix} = [1:0] \quad " \infty "$$

$$(z_1, z_2, z_3) \mapsto (0, 1, \infty) :$$

use matrix $\begin{bmatrix} z_2 - z_3 & -z_1(z_2 - z_3) \\ z_3 - z_1 & -z_3(z_2 - z_1) \end{bmatrix}$

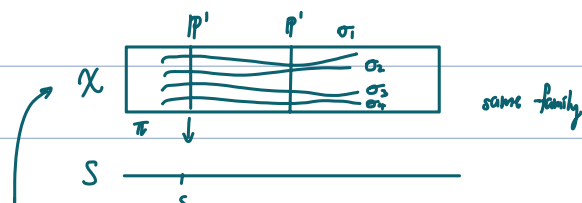
$$\mathcal{A}/\sim \leftrightarrow \{k\text{-pts in } \mathbb{P}^1 \setminus \{0, 1, \infty\}\}.$$

Since the cross-ratio is invariant under Möbius transformations, we can use it to construct moduli spaces.

Example 2.13. Let $\mathcal{A}$ consist of 4 ordered distinct points (p_1, p_2, p_3, p_4) on $\mathbb{P}^1$. We want to classify these quartuples up to the automorphisms of $\mathbb{P}^1$. We recall that the automorphism group of $\mathbb{P}^1$ is the projective linear group PGL_2 , which acts as Möbius transformations. We define our equivalence relation by $(p_1, p_2, p_3, p_4) \sim (q_1, q_2, q_3, q_4)$ if there exists an automorphism $f: \mathbb{P}^1 \rightarrow \mathbb{P}^1$ such that $f(p_i) = q_i$ for $i = 1, \dots, 4$. We recall that for any 3 distinct points (p_1, p_2, p_3) on $\mathbb{P}^1$, there exists a unique Möbius transformation $f \in \text{PGL}_2$ which sends (p_1, p_2, p_3) to $(0, 1, \infty)$ and the cross-ratio of 4 distinct points (p_1, p_2, p_3, p_4) on $\mathbb{P}^1$ is given by $f(p_4) \in \mathbb{P}^1 \setminus \{0, 1, \infty\}$, where f is the unique Möbius transformation that sends (p_1, p_2, p_3) to $(0, 1, \infty)$. Therefore, we see that the set $\mathcal{A}/\sim$ is in bijection with the set of k -points in the quasi-projective variety $\mathbb{P}^1 \setminus \{0, 1, \infty\}$.

In fact, we can naturally speak about families of 4 distinct points on $\mathbb{P}^1$ over a scheme S : this is given by a proper flat morphism $\pi: \mathcal{X} \rightarrow S$ such that the fibres $\pi^{-1}(s) \cong \mathbb{P}^1$ are smooth rational curves and 4 disjoint sections $(\sigma_1, \dots, \sigma_4)$ of π . We say two families $(\pi: \mathcal{X} \rightarrow S, \sigma_1, \dots, \sigma_4)$ and $(\pi': \mathcal{X}' \rightarrow S, \sigma'_1, \dots, \sigma'_4)$ are equivalent over S if there is an isomorphism $f: \mathcal{X} \rightarrow \mathcal{X}'$ over S (i.e. $\pi = \pi' \circ f$) such that $f \circ \sigma_i = \sigma'_i$.

There is a tautological family over the scheme $S = \mathbb{P}^1 \setminus \{0, 1, \infty\}$: let $\pi: \mathbb{P}^1 \setminus \{0, 1, \infty\} \times \mathbb{P}^1 \rightarrow S = \mathbb{P}^1 \setminus \{0, 1, \infty\}$ be the projection map and choose sections $(\sigma_1(s) = 0, \sigma_2(s) = 1, \sigma_3(s) = \infty, \sigma_4(s) = s)$. It turns out that this family over $\mathbb{P}^1 \setminus \{0, 1, \infty\}$ encodes all families parametrised by schemes S (in the language to come, $\mathcal{U}$ is a *universal family* and $\mathbb{P}^1 \setminus \{0, 1, \infty\}$ is a *fine moduli space*).



$\sigma_1, \dots, \sigma_4$: pick points.

(section means $S \rightarrow \mathcal{X}$ s.t. $\pi \circ \sigma_i = \text{id}_S$)

$$\text{family } \mathcal{X} = \mathbb{P}^1 \setminus \{0, 1, \infty\} \times \mathbb{P}^1 \quad (s, t)$$

$$\downarrow \pi \downarrow$$

$$S$$

$$S = \mathbb{P}^1 \setminus \{0, 1, \infty\}$$

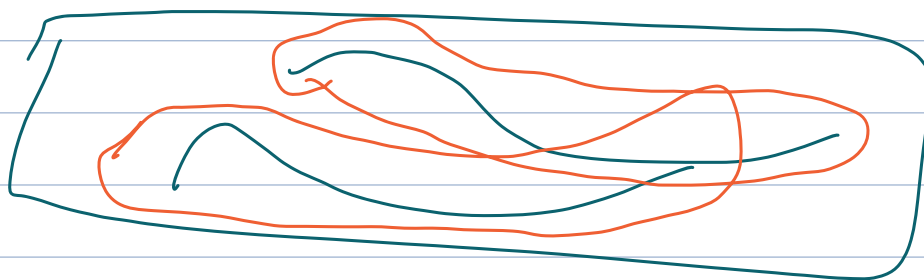
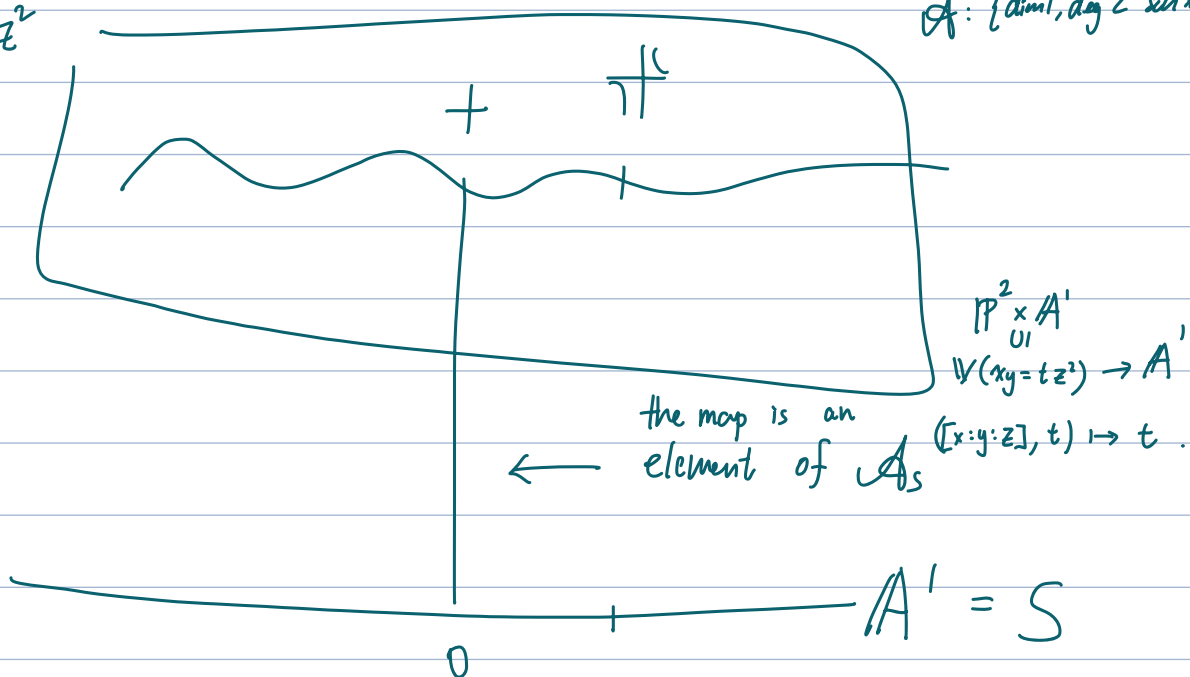
$$\sigma_1(s) = 0, \sigma_2(s) = 1, \sigma_3(s) = \infty, \sigma_4(s) = s.$$

Later: this is a fine moduli space.

Idea Plus

$$xy = tz^2$$

$\mathcal{A}: \{\text{dim 1, deg 2 sch in } \mathbb{P}^2\}$



$\mathcal{A}$



T

$\downarrow$

S

$\downarrow$

$\text{Spec } k$

Exercise time!

Exercise 2.6. Show that if there is a natural isomorphism $h_C \rightarrow h'_C$, then there is a canonical isomorphism $C \rightarrow C'$.

By Yoneda's lemma,

$$\{\text{natural transformations } \eta: h_C \rightarrow h_{C'}\} \xleftrightarrow{\text{bij.}} h_{C'}(C) = \text{Hom}(C, C').$$

$$\eta \mapsto \eta_C(\text{id}_C)$$

It remains to show if η is an iso, then $\eta_C(\text{id}_C)$ is also an iso.

Since η is an iso, $\exists$ inverse $\theta: h_{C'} \rightarrow h_C$, i.e. $\theta\eta = \text{id}_{h_{C'}}$, $\theta\eta = \text{id}_{h_C}$.

Let $g = \theta_{C'}(\text{id}_{C'}) \in \text{Hom}(C', C)$.

Claim: g is invm of f , i.e. $g \circ f = \text{id}_C$, $f \circ g = \text{id}_{C'}$.

① $g \circ f$:

Diagram for f and θ :

$$f: C \rightarrow C'$$

$$\theta: h_{C'} \rightarrow h_C$$

h_C contravariant, so vertical arrows are inverted.

$$\begin{array}{ccc} h_{C'}(C) & \xrightarrow{\theta_C} & h_C(C) \\ \uparrow h_{C'}(f) & & \uparrow h_C(f) \\ h_{C'}(C') & \xrightarrow{\theta_{C'}} & h_C(C') \end{array}$$

$$\text{Hom}(C, C') \xrightarrow{\theta_C} \text{Hom}(C, C)$$

$$\begin{array}{ccc} \text{Hom}(C, C') & \xrightarrow{\theta_C} & \text{Hom}(C, C) \\ \uparrow & & \uparrow \\ \text{Hom}(C', C') & \xrightarrow{\theta_{C'}} & \text{Hom}(C', C) \end{array}$$

i.e. $\frac{\text{sth in } \text{Hom}(C', C')}{\text{Hom}(C', C')} \circ f = h_{C'}(f)$ $\quad$ $h_C(f) = \frac{\text{sth in } \text{Hom}(C', C)}{\text{Hom}(C', C)} \circ f$

$$\begin{array}{l}
 \text{id}_{C'} \circ f \longrightarrow \Theta_C(\text{id}_{C'} \circ f) = \Theta_C(f) \\
 \text{id}_{C'} \qquad \qquad \qquad = \Theta_C(\eta_C(\text{id}_C)) = \text{id}_C.
 \end{array}$$

$$\begin{array}{l}
 \qquad \qquad \qquad g \circ f \\
 \qquad \qquad \qquad \uparrow \\
 \text{id}_C \longrightarrow \Theta_{C'}(\text{id}_C) = g
 \end{array}$$

$$\therefore g \circ f = \text{id}_C$$

similar for the other direction.

Exercise 2.14. Define an analogous notion for families of n ordered distinct points on $\mathbb{P}^1$ and let the corresponding moduli functor be denoted $\mathcal{M}_{0,n}$ (this is the moduli functor of n ordered distinct points on the curve $\mathbb{P}^1$ of genus 0). For $n = 3$, show that $\mathcal{M}_{0,3}(\text{Spec } k)$ is a single element set and so is in bijection with the set of k -points of $\text{Spec } k$. Furthermore, show there is a tautological family over $\text{Spec } k$.

The functor $\mathcal{M}_{0,n} : \text{Sch}^{\text{op}} \rightarrow \text{Sets}$

should map a scheme S to the set of all tuples

$$S \mapsto \mathcal{M}_{0,n}(S) = \left\{ \begin{array}{l} \text{families } \pi : X \rightarrow S, \\ \text{sections } \sigma_1, \sigma_2, \dots, \sigma_n \end{array} \right\} / \sim$$

s.t.

- $X_S = \pi^{-1}(s) \cong \mathbb{P}^1$ (smooth connected algebraic curve of genus 0)
- σ_i are sections of π , i.e. $\pi \circ \sigma_i = \text{id}_S \quad \forall i$.
- Images of σ_i should be different : $\sigma_i(s) \neq \sigma_j(s) \quad \forall s \in S$.
- $X \sim X'$ if $\exists f : X \xrightarrow{\sim} X'$ s.t. $f \circ \sigma_i = \sigma'_i \quad \forall i$.

Act on morphisms :

If $g : T \rightarrow S$, $\mathcal{M}_{0,n}(g) : \mathcal{M}_{0,n}(S) \rightarrow \mathcal{M}_{0,n}(T)$ by pullback.
 (family) $X \mapsto g^*X$

Again, we can use an automorphism to make $\sigma_1(s) = 0, \sigma_2(s) = 1, \sigma_3(s) = \infty$.

But no restriction on the other points.

$\therefore \mathcal{M}_{0,n}$ is representable by
 (sometimes just write $\cong$)

$$S \cong (\mathbb{P}^1 \setminus \{0, 1, \infty\})^{n-3} \setminus \Delta$$

where Δ represent all tuples where any two of the points match.

For $n = 3$, $\mathcal{M}_{0,n} \cong S \cong (\mathbb{P}^1 \setminus \{0, 1, \infty\})^0 = \{\cdot\}$ (singleton).

Taut. family over $\text{Spec } k$: $(\pi : \mathbb{P}^1, \sigma_1, \sigma_2, \sigma_3)$ (taut. : every elt in this family)
 $\downarrow$
 $\text{Spec}(k)$